Reparametrization Invariance in HQET Amplitudes

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NORTHWESTERN UNIVERSITY

Outline

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- Why HQET?

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- Reparametrization invariance

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- HQET spinors

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- Fixing 3-pt amplitudes via RPI

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- Fixing 3-pt amplitudes via RPI
- Off-shell vertices in on-shell formalism

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- HQET and NRQCD

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Heavy quark field [Manohar, Wise]:

$$\psi_v(x) \equiv e^{imv \cdot x} \frac{1 + \not{v}}{2} \psi(x)$$

Heavy Quark Field-2

Field redefinition:

$$\psi_v(x) \equiv e^{imv\cdot x}\frac{1+\not{v}}{2}\psi(x)$$

$$H_v(x) \equiv e^{imv\cdot x}\frac{1-\not{v}}{2}\psi(x)$$

with $\not{v}\psi_v = \psi_v$ and $\not{v}H_v = -H_v$.

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where $D_\perp^\mu = D^\mu - v^\mu(v\cdot D)$.

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where $D_\perp^\mu = D^\mu - v^\mu(v\cdot D)$.

$$\psi_v \xrightarrow{\quad p \quad} \approx \frac{1}{v\cdot k + i\epsilon}\frac{1+\not{v}}{2}$$

HQET Lagrangian

Integrate out $H_v(x)$,

$$\begin{aligned}\mathcal{L}_{\mathsf{I,QCD}} &= \bar{\psi}_v(iv \cdot D)\psi_v - \bar{H}_v(iv \cdot D + 2m)H_v \\ &\quad + \bar{\psi}_v \; i\not{D}_\perp H_v + \bar{H}_v \; i\not{D}_\perp \psi_v, \\ \to \mathcal{L}_{\mathsf{HQET}} &= \bar{\psi}_v \left(iv \cdot D + i\not{D}_\perp \frac{1}{2m + iv \cdot D} i\not{D}_\perp \right) \psi_v.\end{aligned}$$

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Expansion in $1/m$:

$$\mathcal{L}_{\mathsf{HQET}} = \mathcal{L}_0 + \frac{1}{m}\mathcal{L}_1 + \cdots$$

Reparametrization Invariance

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Reparametrization invariance is a non-linearly realized Lorentz symmetry, which relates different orders!

HQET Lagrangian at Order $1/m^3$-1

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To order $1/m^3$ [Manohar, 1997]:

$$\mathcal{L}_{\text{tree}} = \bar{\psi}_v \Big\{ iv \cdot D - \frac{D_\perp^2}{2m} + \frac{D_\perp^4}{8m^3} - g\frac{\sigma_{\mu\nu}G^{\mu\nu}}{4m} - g\frac{v_\mu \left(D_{\perp\nu}G^{\mu\nu}\right)}{8m^2} + ig\frac{v_\mu \sigma_{\nu\rho}\left\{D_\perp^\nu, G^{\mu\rho}\right\}}{8m^2} + g\frac{\left\{D_\perp^2, \sigma_{\mu\nu}G^{\mu\nu}\right\}}{16m^3} + g^2\frac{G^2}{16m^3} - ig^2\frac{\sigma_{\mu\nu}[G^\mu_\rho, G^{\rho\nu}]}{16m^3} + \mathcal{O}(\frac{1}{m^4}) \Big\} \psi_v$$

HQET Lagrangian at Order $1/m^3$-2

1-loop correction:

$$
\begin{aligned}
\mathcal{L}_{\text{1-loop}} =& \sum_i c_i \hat{O}_i + \mathcal{O}(g^2) \\
=& \bar{\psi}_v \Bigg\{ iv\cdot D - c_2 \frac{D_\perp^2}{2m} + c_4 \frac{D_\perp^4}{8m^3} - c_F g \frac{\sigma_{\mu\nu} G^{\mu\nu}}{4m} \\
& - c_D g \frac{v_\mu \left(D_{\perp\nu} G^{\mu\nu} \right)}{8m^2} + i c_S g \frac{v_\mu \sigma_{\nu\rho} \left\{ D_\perp^\nu, G^{\mu\rho} \right\}}{8m^2} \\
& + c_{W_1} g \frac{\left\{ D_\perp^2, \sigma_{\mu\nu} G^{\mu\nu} \right\}}{16m^3} - c_{W_2} g \frac{D_{\perp\rho} \sigma_{\mu\nu} G^{\mu\nu} D_\perp^\rho}{8m^3} \\
& + c_{p'p} g \frac{\sigma_{\mu\nu} \left(D_{\perp\rho} G^{\rho\mu} D_\perp^\nu + D_\perp^\nu G^{\rho\mu} D_{\perp\rho} - D_{\perp\rho} G^{\mu\nu} D_\perp^\rho \right)}{8m^3} \\
& - i c_M g \frac{D_{\perp\mu} \left(D_{\perp\nu} G^{\mu\nu} \right) + \left(D_{\perp\nu} G^{\mu\nu} \right) D_{\perp\mu}}{8m^3} + \mathcal{O}(\frac{1}{m^4}) \Bigg\} \psi_v
\end{aligned}
$$

HQET Lagrangian at Order $1/m^3$-3

Comparing to QCD:

$$\begin{aligned} c_2 &= c_4 = 1 & c_F &= 1 + F_2 \\ c_{p'p} &= F_2 & c_S &= 1 + 2F_2 \\ c_D &= 1 + 2F_2 + 8F_1' & c_M &= -\frac{1}{2}F_2 - 4F_1' \\ c_{W_1} &= 1 + \frac{1}{2}F_2 + 4F_1' + 4F_2' & c_{W_2} &= \frac{1}{2}F_2 + 4F_1' + 4F_2', \end{aligned}$$

where $F_2 \equiv F_2(0)$, $F_i' \equiv m^2 \frac{dF_i(q^2)}{dq^2}|_{q^2=0}$, and
$V^a_{3,\mu} = -igT^a \bar{u}(p_2) \left[F_1(q^2)\gamma_\mu + iF_2(q^2)\frac{\sigma_{\mu\nu}q^\nu}{2m} \right] u(p_1)$.

HQET Lagrangian at Order $1/m^3$-4

RPI operartors:

$$iv \cdot D + \hat{O}_2 + \hat{O}_4 + \hat{O}_F + \hat{O}_D + \hat{O}_S + \hat{O}_{W_1}$$

$$2\hat{O}_F + 4\hat{O}_D + 4\hat{O}_S + \hat{O}_{W_1} + \hat{O}_{W_2} + 2\hat{O}_{p'p} - \hat{O}_M$$

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Wilson coefficients of different operators are related by reparametrization invariance!

HQET Lagrangian at Order $1/m^3$-4

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Wilson coefficients of different operators are related by reparametrization invariance!

RPI constraints:

$$c_2=1 \qquad c_4=1$$

$$c_S=2c_F-1 \qquad c_{W_2}=c_{W_1}-1$$

$$c_{p'p}=c_F-1 \qquad 2c_M=c_F-c_D$$

Massive On-Shell Spinors-1

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For massive fermions [Arkani-Hamed, 2021]:

$$u^I(p) \equiv \begin{pmatrix} \lambda^I_\alpha \\ \tilde{\lambda}^{I,\dot{\alpha}} \end{pmatrix} \equiv \begin{pmatrix} |p^I\rangle_\alpha \\ |p^I]^{\dot{\alpha}} \end{pmatrix}$$

$$\bar{u}^I(p) \equiv \begin{pmatrix} -\lambda^{I,\alpha}, & \tilde{\lambda}^I_{\dot{\alpha}} \end{pmatrix} \equiv \begin{pmatrix} -\langle p^I|^\alpha, & [p^I|_{\dot{\alpha}} \end{pmatrix}$$

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3-pt with 2 massive and 1 massless particles:

$$\begin{aligned} & M_3^{h,(I_1\cdots I_{2s_1})(J_1\cdots J_{2s_2})} \\ = \; & M^h_{3,(\alpha_1\cdots\alpha_{2s_1})(\beta_1\cdots\beta_{2s_2})} \lambda_1^{I_1,\alpha_1} \cdots \lambda_1^{I_{2s_1},\alpha_{2s_1}} \lambda_2^{J_1,\beta_1} \cdots \lambda_2^{J_{2s_2},\beta_{2s_2}} \end{aligned}$$

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$$M^h_{3,(\alpha_1\cdots\alpha_{2s_1})(\beta_1\cdots\beta_{2s_2})} = \sum_{i,j} c_{ij}\left(\eta_1^i\eta_2^{2s_1+2s_2-i}\right)^{(j)}_{(\alpha_1\cdots\alpha_{2s_1})(\beta_1\cdots\beta_{2s_2})}$$

Massive On-Shell Spinors-2

$$M^h_{3,(\alpha_1\cdots\alpha_{2s_1})(\beta_1\cdots\beta_{2s_2})} = \sum_{i,j} c_{ij} \left(\eta_1^i \eta_2^{2s_1+2s_2-i}\right)^{(j)}_{(\alpha_1\cdots\alpha_{2s_1})(\beta_1\cdots\beta_{2s_2})}$$

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$m_1 \neq m_2$:

$$\left(\eta_{1\alpha}, \quad \eta_{2\alpha}\right) = \left(|q\rangle_\alpha, \quad \frac{p_{1\alpha\dot{\alpha}}}{m_1}|q]^{\dot{\alpha}}\right)$$

Massive On-Shell Spinors-2

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Matching the helicity:

$$M^h_{3,(\alpha_1\cdots\alpha_{2s_1})(\beta_1\cdots\beta_{2s_2})} = \sum_{j} c_{j} \left(\eta_1^{s_1+s_2-h} \eta_2^{s_1+s_2+h}\right)^{(j)}_{(\alpha_1\cdots\alpha_{2s_1})(\beta_1\cdots\beta_{2s_2})}$$

Massive On-Shell Spinor-3

$m_1 = m_2 = m$:

$$\eta_1^\alpha \eta_{2\alpha} = \langle q | \frac{p_1}{m} | q] = \frac{p_1 \cdot q}{m} = 0$$

η_1 and η_2 do not span the space!

Massive On-Shell Spinor-3

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η_1 and η_2 do not span the space!

x-factor:

$$[q|p_1 = mx\langle q|$$

$$x = \frac{[q|p_1|\zeta\rangle}{m\langle q\zeta\rangle} = \sqrt{2}\,\frac{p_1 \cdot \epsilon^+}{m}$$

Massive On-Shell Spinor-4

With the anti-symmetric tensor $\varepsilon_{\alpha\beta}$,

$$M^{h}_{3,(\alpha_1\cdots\alpha_{2s_1})(\beta_1\cdots\beta_{2s_2})} = \sum_{i=|s_1-s_2|}^{s_1+s_2} c_i x^{h+i} \left(\eta_1^{2i}\varepsilon^{s_1+s_2-i}\right)_{(\alpha_1\cdots\alpha_{2s_1})(\beta_1\cdots\beta_{2s_2})}$$

Massive On-Shell Spinor-4

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2 equally massive spin-$\frac{1}{2}$ and 1 massless spin-1:

$$M_3^{+1,IJ} = c_1 x\langle 2^I 1^J\rangle + c_2 \frac{x^2}{m}\langle 2^I q\rangle\langle q 1^J\rangle$$

HQET Spinors

HQET Spinors

Define heavy quark spinors [Aoude, 2020]:

$$u_v(p) \equiv \frac{1+\not{v}}{2} u(p) = \left(1 - \frac{\not{k}}{2m}\right) u(p)$$

HQET Spinors

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$$u_v(p) \equiv \frac{1+\not{v}}{2} u(p) = \left(1 - \frac{\not{k}}{2m}\right) u(p)$$

$$v|p_v] = |p_v\rangle, \quad \bar{v}|p_v\rangle = |p_v]$$

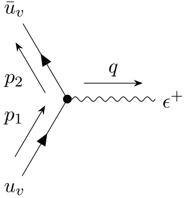
Conversion rules:

$$|p_v\rangle = \left(1 + \frac{v\bar{k}}{2m}\right)|p\rangle, \qquad |p\rangle = \frac{1}{1-\alpha}\left(1 + \frac{k\bar{v}}{2m}\right)|p_v\rangle,$$

$$|p_v] = \left(1 + \frac{\bar{v}k}{2m}\right)|p], \qquad |p] = \frac{1}{1-\alpha}\left(1 + \frac{\bar{k}v}{2m}\right)|p_v],$$

where $\alpha = \frac{k^2}{4m^2}$.

3-pt Vertex of Two Quarks and One Photon



$$= x\langle 2_v|M_{2\times 2}|1_v\rangle$$

$$= x\left(\tilde{c}_1\langle 2_v 1_v\rangle + \tilde{c}_2\langle 2_v|\frac{k_1}{m}|1_v] + \tilde{c}_3[2_v|\frac{\bar{k}_2}{m}|1_v\rangle + \tilde{c}_4\langle 2_v|\frac{k_2\bar{k}_1}{m^2}|1_v\rangle\right),$$

where $p_i = mv + k_i$, $q = p_1 - p_2 = q_1 - q_2$ and $\tilde{c}_i = \tilde{c}_i(\alpha_1, \alpha_2)$.

Reparametrization Invariance in On-Shell Variables-1

Reparametrization Invariance in On-Shell Variables-1

$$p = mv + k = mw + k', \quad w = v + \frac{\ell}{m}, \quad k' = k - \ell$$

Conversion from w to v:

$$|p_w\rangle = \left[1 + \frac{1}{1-\alpha}\frac{\ell}{2m}\left(\bar{v} + \frac{\bar{k}}{2m}\right)\right]|p_v\rangle$$
$$|p_w] = \left[1 + \frac{1}{1-\alpha}\frac{\bar{\ell}}{2m}\left(v + \frac{k}{2m}\right)\right]|p_v]$$

RPI relates different orders again!

Reparametrization Invariance in On-Shell Variables-2

Starting with $p_i = mw + k'_i$:

$$\frac{1}{x}M_{3,w} = \tilde{c}'_1\langle 2_w 1_w\rangle + \tilde{c}'_2\langle 2_w|\frac{k'_1}{m}|1_w] + \tilde{c}'_3[2_w|\frac{\bar{k}'_2}{m}|1_w\rangle + \tilde{c}'_4\langle 2_w|\frac{k'_2\bar{k}'_1}{m^2}|1_w\rangle$$
$$= \bar{c}_1\langle 2_v 1_v\rangle + \bar{c}_2\langle 2_v|\frac{k_1}{m}|1_v] + \bar{c}_3[2_v|\frac{\bar{k}_2}{m}|1_v\rangle + \bar{c}_4\langle 2_v|\frac{k_2\bar{k}_1}{m^2}|1_v\rangle$$
$$+ \bar{c}_5\langle 2_v|\frac{\ell\bar{k}_1}{m^2}|1_v\rangle + \bar{c}_6\langle 2_v|\frac{k_2\bar{\ell}}{m^2}|1_v\rangle + \bar{c}_7\langle 2_v|\frac{\ell}{m}|1_v]$$
$$+ \bar{c}_8\langle 2_v|\frac{k_2\bar{\ell}k_1}{m^3}|1_v]$$
$$\frac{1}{x}M_{3,v} = \tilde{c}_1\langle 2_v 1_v\rangle + \tilde{c}_2\langle 2_v|\frac{k_1}{m}|1_v] + \tilde{c}_3[2_v|\frac{\bar{k}_2}{m}|1_v\rangle + \tilde{c}_4\langle 2_v|\frac{k_2\bar{k}_1}{m^2}|1_v\rangle,$$

where $\ell = k_i - k'_i = m(w-v)$.

Reparametrization Invariance in On-Shell Variables-2

Starting with $p_i = mw + k_i'$:

$$
\begin{aligned}
\frac{1}{x}M_{3,w} &= \tilde{c}_1'\langle 2_w 1_w\rangle + \tilde{c}_2'\langle 2_w|\frac{k_1'}{m}|1_w] + \tilde{c}_3'[2_w|\frac{\bar{k}_2'}{m}|1_w\rangle + \tilde{c}_4'\langle 2_w|\frac{k_2'\bar{k}_1'}{m^2}|1_w\rangle \\
&= \bar{c}_1\langle 2_v 1_v\rangle + \bar{c}_2\langle 2_v|\frac{k_1}{m}|1_v] + \bar{c}_3[2_v|\frac{\bar{k}_2}{m}|1_v\rangle + \bar{c}_4\langle 2_v|\frac{k_2\bar{k}_1}{m^2}|1_v\rangle \\
&\quad + \bar{c}_5\langle 2_v|\frac{\ell\bar{k}_1}{m^2}|1_v\rangle + \bar{c}_6\langle 2_v|\frac{k_2\bar{\ell}}{m^2}|1_v\rangle + \bar{c}_7\langle 2_v|\frac{\ell}{m}|1_v] \\
&\quad + \bar{c}_8\langle 2_v|\frac{k_2\bar{\ell}k_1}{m^3}|1_v] \\
\frac{1}{x}M_{3,v} &= \tilde{c}_1\langle 2_v 1_v\rangle + \tilde{c}_2\langle 2_v|\frac{k_1}{m}|1_v] + \tilde{c}_3[2_v|\frac{\bar{k}_2}{m}|1_v\rangle + \tilde{c}_4\langle 2_v|\frac{k_2\bar{k}_1}{m^2}|1_v\rangle,
\end{aligned}
$$

where $\ell = k_i - k_i' = m(w - v)$.

Reparametrization invariance: $\bar{c}_i = \tilde{c}_i$

Reparametrization Invariance in On-Shell Variables-3

RPI constraints:

$$\frac{\alpha'_1 - \alpha'_2}{2} c'_1 = (1 - 2\alpha'_1 + 2\alpha'^2_1 - \alpha'_2) c'_2 + (1 - 2\alpha'_2 + 2\alpha'^2_2 - \alpha'_1) c'_3$$

$$2(\alpha'_1 - \alpha'_2) c'_4 = (1 - \alpha'_1) c'_2 + (1 - \alpha'_2) c'_3,$$

where $\alpha'_i = \frac{k'^2_i}{4m^2}$.

Reparametrization Invariance in On-Shell Variables-4

RPI amplitudes:

$$
\begin{aligned}
M_{3,v,\mathsf{RPI}} = A \Bigg[& \langle 2_v 1_v \rangle + \left(\frac{1}{2} - \alpha_2\right) \langle 2_v | \frac{k_1}{m} | 1_v] \\
& + \left(-\frac{1}{2} + \alpha_1\right) [2_v | \frac{\bar{k}_2}{m} | 1_v \rangle + \frac{1}{4} \langle 2_v | \frac{k_2 \bar{k}_1}{m^2} | 1_v \rangle \Bigg] \\
& + B \Bigg[2 (\alpha_1 + \alpha_2) \langle 2_v 1_v \rangle + (1 - \alpha_2) \langle 2_v | \frac{k_1}{m} | 1_v] \\
& - (1 - \alpha_1) [2_v | \frac{\bar{k}_2}{m} | 1_v \rangle \Bigg] \\
\approx\ & (1 - \alpha_1)(1 - \alpha_2) \left(A \langle 21 \rangle + B \frac{x}{m} \langle 2q \rangle \langle q1 \rangle \right)
\end{aligned}
$$

Reparametrization Invariance in On-Shell Variables-4

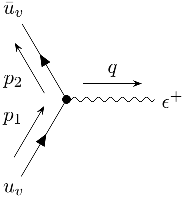
RPI amplitudes:

$$
\begin{aligned}
M_{3,v,\mathsf{RPI}} = A \Bigg[& \langle 2_v 1_v \rangle + \left(\frac{1}{2} - \alpha_2\right) \langle 2_v | \frac{k_1}{m} | 1_v] \\
& + \left(-\frac{1}{2} + \alpha_1\right) [2_v | \frac{\bar{k}_2}{m} | 1_v \rangle + \frac{1}{4} \langle 2_v | \frac{k_2 \bar{k}_1}{m^2} | 1_v \rangle \Bigg] \\
& + B \Bigg[2 \left(\alpha_1 + \alpha_2\right) \langle 2_v 1_v \rangle + \left(1 - \alpha_2\right) \langle 2_v | \frac{k_1}{m} | 1_v] \\
& - \left(1 - \alpha_1\right) [2_v | \frac{\bar{k}_2}{m} | 1_v \rangle \Bigg] \\
\approx \; & \left(1 - \alpha_1\right)\left(1 - \alpha_2\right) \left(A \langle 21 \rangle + B \frac{x}{m} \langle 2q \rangle \langle q1 \rangle \right)
\end{aligned}
$$

Restore QCD again!

Comparison between Lagrangian and On-Shell Formalism-1

The 3-pt amplitude in Lagrangian formalism:



$$= f(c_i, \alpha_j, m) = \text{ RPI}$$

Comparison between … hell Formalism-1

The 3-pt …

$$e\bar{Q}_v\Big\{-v\cdot A-\frac{ic_2}{2m}(\partial_\perp\cdot A_\perp+A_\perp\cdot\partial_\perp)+\frac{ic_4}{8m^3}\big[\partial_\perp^2(\partial_\perp\cdot A_\perp)+\partial_\perp^2(A_\perp\cdot\partial_\perp)+(\partial_\perp\cdot A_\perp)\partial_\perp^2+(A_\perp\cdot\partial_\perp)\ldots$$
$$-\frac{c_F}{2m}\sigma_{\alpha\beta}[\partial^\alpha A^\beta]-\frac{c_D}{8m^2}v^\alpha\big[\partial_\perp^\beta(\partial_\alpha A_\beta-\partial_\beta A_\alpha)\big]+\frac{ic_S}{8m^2}v_\lambda\sigma_{\alpha\beta}\{\partial_\perp^\alpha,([\partial^\lambda A^\beta]-[\partial^\beta A^\lambda])\}$$
$$+\frac{c_{W1}}{8m^3}\sigma_{\alpha\beta}\{\partial_\perp^2,[\partial^\alpha A^\beta]\}-\frac{c_{W2}}{4m^3}\sigma_{\alpha\beta}\partial_\perp^\lambda[\partial^\alpha A^\beta]\partial_{\perp\lambda}$$
$$+\frac{c_{p'p}}{8m^3}\sigma^{\alpha\beta}\big(\partial_\perp^\lambda[\partial_\lambda A_\alpha]\partial_{\perp\beta}-\partial_\perp^\lambda[\partial_\alpha A_\lambda]\partial_{\perp\beta}+\partial_{\perp\beta}[\partial_\lambda A_\alpha]\partial_\perp^\lambda-\partial_{\perp\beta}[\partial_\alpha A_\lambda]\partial_\perp^\lambda-2\partial_\perp^\lambda[\partial_\alpha A_\beta]\partial_{\perp\lambda}\big)$$
$$-\frac{ic_M}{8m^3}\big(\partial_{\perp\alpha}[\partial_{\perp\beta}\partial^\alpha A^\beta]-\partial_{\perp\alpha}[\partial_{\perp\beta}\partial^\beta A^\alpha]+[\partial_{\perp\beta}\partial^\alpha A^\beta]\partial_{\perp\alpha}-[\partial_{\perp\beta}\partial^\beta A^\alpha]\partial_{\perp\alpha}\big)\Big\}Q_v. \tag{35}$$

Therefore, the 3-pt vertex is

$$e\bar{u}_{2v}\Big\{-v\cdot\epsilon-\frac{c_2}{2m}(k_{1\perp}+k_{2\perp})\cdot\epsilon-\frac{c_4}{8m^3}(k_{1\perp}^2+k_{2\perp}^2)(k_{1\perp}+k_{2\perp})\cdot\epsilon-\frac{ic_F}{2m}\sigma_{\alpha\beta}q^\alpha\epsilon^\beta+c_D\cdot 0$$
$$+\frac{ic_S}{8m^2}\sigma_{\alpha\beta}v_\lambda(q^\lambda\epsilon^\beta-q^\beta\epsilon^\lambda)(k_{1\perp}^\alpha+k_{2\perp}^\alpha)-\frac{ic_{W1}}{8m^3}\sigma_{\alpha\beta}(k_{1\perp}^2+k_{2\perp}^2)q^\alpha\epsilon^\beta+\frac{ic_{W2}}{4m^3}\sigma_{\alpha\beta}(k_{1\perp}\cdot k_{2\perp})q^\alpha\epsilon^\beta$$
$$-\frac{ic_{p'p}}{8m^3}\sigma_{\alpha\beta}\big((q\cdot k_{1\perp})\epsilon^\alpha k_{2\perp}^\beta-(\epsilon\cdot k_{1\perp})q^\alpha k_{2\perp}^\beta+(q\cdot k_{2\perp})\epsilon^\alpha k_{1\perp}^\beta-(\epsilon\cdot k_{2\perp})q^\alpha k_{1\perp}^\beta-2(k_{1\perp}\cdot k_{2\perp})q^\alpha\epsilon^\beta\big)$$
$$+\frac{c_M}{8m^3}\big((v\cdot q)^2\epsilon\cdot(k_{1\perp}+k_{2\perp})-(v\cdot q)(v\cdot\epsilon)q\cdot(k_{1\perp}+k_{2\perp})\big)\Big\}u_{1v}. \tag{36}$$

Assume the helicity of photon is +1. In the on-shell HQET variables, it is

$$\Big\{-\frac{y_\zeta}{\sqrt{2}}-\frac{1}{\sqrt{2}}\Big(c_2+c_4\Big[\frac{k_1^2}{4m^2}\Big(1-\frac{k_1^2}{4m^2}\Big)+\frac{k_2^2}{4m^2}\Big(1-\frac{k_2^2}{4m^2}\Big)\Big]\Big)\Big[x-y_\zeta\Big(1-\frac{k_1^2}{4m^2}-\frac{k_2^2}{4m^2}\Big)\Big]$$
$$+\frac{c_M}{\sqrt{2}}x\Big(\frac{k_1^2}{4m^2}-\frac{k_2^2}{4m^2}\Big)^2\Big\}\cdot 2[2_v1_v]$$
$$+\frac{[2_vq][q1_v]}{\sqrt{2}m}\Big(c_F+c_{W1}\Big[\frac{k_1^2}{4m^2}\Big(1-\frac{k_1^2}{4m^2}\Big)+\frac{k_2^2}{4m^2}\Big(1-\frac{k_2^2}{4m^2}\Big)\Big]-c_{W2}\Big[\frac{k_1^2}{4m^2}\Big(1-\frac{k_2^2}{4m^2}\Big)+\frac{k_2^2}{4m^2}\Big(1-\frac{k_1^2}{4m^2}\Big)\Big]\Big)$$
$$+\ldots\Big[\frac{x}{2}\big(\langle 2_vq\rangle[q1_v]+[2_vq]\langle q1_v\rangle\big)-\Big(1-\frac{k_1^2}{4m^2}-\frac{k_2^2}{4m^2}\Big)[2_vq][q1_v]\Big]$$
$$\ldots\langle 2_vq\rangle\langle q1_v\rangle-x\Big(1-\frac{k_1^2}{4m^2}-\frac{k_2^2}{4m^2}\Big)\big([2_vq\rangle\langle q1_v\rangle+\langle 2_vq\rangle[q1_v\rangle\big)+\Big(1-2\frac{k_1^2+k_2^2}{m^2}+\frac{k_1^2k_2^2}{m^4}\Big)[2_vq][q1_v]\Big]. \tag{37}$$

Compar... Shell Formalism-1

T

$$
\begin{aligned}
&\sqrt{2}(1+\alpha_1+\alpha_2)x\langle 2_v 1_v\rangle \\
&+\left[\left(c_F-\frac{c_S}{2}+\frac{c_{p'p}}{4}\right)+\left(c_{W_1}-c_{W_2}+\frac{c_S}{2}-2c_{p'p}\right)(\alpha_1+\alpha_2)\right]\frac{[2_v q][q 1_v]}{\sqrt{2}m} \\
&+[(c_S-c_{p'p})+c_{p'p}(\alpha_1+\alpha_2)]\frac{x}{4\sqrt{2}m}\left(\langle 2_v q\rangle[q 1_v]+[2_v q]\langle q 1_v\rangle\right) \\
&+\frac{c_{p'p}}{4\sqrt{2}m}x^2\langle 2_v q\rangle\langle q 1_v\rangle+\mathcal{O}(1/m^4) \qquad (35)\\
=&\sqrt{2}(1+\alpha_1+\alpha_2)x\langle 2_v 1_v\rangle \\
&+\left[\left(c_F-\frac{c_S}{2}+\frac{c_{p'p}}{4}\right)+\left(c_{W_1}-c_{W_2}+\frac{c_S}{2}-2c_{p'p}\right)(\alpha_1+\alpha_2)\right] \\
&\times\frac{x}{\sqrt{2}}\left[(1-4\alpha_2)\langle 2_v|\frac{k_1}{m}|1_v]+(-1+4\alpha_1)[2_v|\frac{k_2}{m}|1_v\rangle+\langle 2_v|\frac{k_2k_1}{m^2}|1_v\rangle\right] \\
&+[(c_S-c_{p'p})+c_{p'p}(\alpha_1+\alpha_2)] \\
&\times\frac{x}{2\sqrt{2}}\left(\langle 2_v|\frac{k_1}{m}|1_v]-[2_v|\frac{k_2}{m}|1_v\rangle+2(\alpha_1+\alpha_2)\langle 2_v 1_v\rangle\right) \\
&+\frac{c_{p'p}}{4\sqrt{2}}x\left(4(\alpha_1+\alpha_2)\langle 2_v 1_v\rangle+\langle 2_v|\frac{k_1}{m}|1_v]-[2_v|\frac{k_2}{m}|1_v\rangle-\langle 2_v|\frac{k_2k_1}{m^2}|1_v\rangle\right)+\mathcal{O}(1/m^4) \\
=&\left[1+\left(1+\frac{c_S}{2}\right)(\alpha_1+\alpha_2)\right]\sqrt{2}x\langle 2_v 1_v\rangle \\
&+\left[c_F+\left(c_{W_1}-c_{W_2}+\frac{c_S}{2}-\frac{3}{2}c_{p'p}\right)(\alpha_1+\alpha_2)\right]\frac{x}{\sqrt{2}}\left(\langle 2_v|\frac{k_1}{m}|1_v]-[2_v|\frac{k_2}{m}|1_v\rangle\right) \\
&-\left(c_F-\frac{c_S}{2}+\frac{c_{p'p}}{4}\right)2\sqrt{2}x\left(\alpha_2\langle 2_v|\frac{k_1}{m}|1_v]-\alpha_1[2_v|\frac{k_2}{m}|1_v\rangle\right) \\
&+\left(c_F-\frac{c_S}{2}\right)\frac{x}{\sqrt{2}}\langle 2_v|\frac{k_2k_1}{m^2}|1_v\rangle+\mathcal{O}(1/m^4)
\end{aligned}
$$

To satisfy the RPI constraints in the previous section, we need

$$
\begin{aligned}
3c_F-2c_S+c_{p'p}&=1 \\
c_F-c_S+c_{p'p}&=0,
\end{aligned}
$$

Comparison between Lagrangian and On-Shell Formalism-2

$$M_3 = -\tilde{c}_\zeta \frac{y_\zeta}{\sqrt{2}}$$

$$+ \sqrt{2}x\left(\tilde{c}_1\langle 2_v 1_v\rangle + \tilde{c}_2\langle 2_v|\frac{k_1}{m}|2_v] + \tilde{c}_3[2_v|\frac{\bar{k}_2}{m}|1_v\rangle + \tilde{c}_4\langle 2_v|\frac{k_2\bar{k}_1}{m^2}|1_v\rangle\right),$$

where $y_\zeta = \frac{[q|v|\zeta\rangle}{\langle q\zeta\rangle}$ is ζ-dependent!

Comparison between Lagrangian and On-Shell Formalism-3

$$
\begin{aligned}
\tilde{c}_\zeta &= (1-c_2)+(c_2-c_4)(\alpha_1+\alpha_2) \\
\tilde{c}_1 &= 1+\left(1+\frac{c_S}{2}\right)(\alpha_1+\alpha_2) \\
\tilde{c}_2 &= \frac{1}{2}\left[c_F+\left(c_{W_1}-c_{W_2}+\frac{c_S}{2}-\frac{3}{2}c_{p'p}\right)(\alpha_1+\alpha_2)\right] \\
&\quad -2\left(c_F-\frac{c_S}{2}+\frac{c_{p'p}}{4}\right)\alpha_2 \\
\tilde{c}_3 &= -\frac{1}{2}\left[c_F+\left(c_{W_1}-c_{W_2}+\frac{c_S}{2}-\frac{3}{2}c_{p'p}\right)(\alpha_1+\alpha_2)\right] \\
&\quad +2\left(c_F-\frac{c_S}{2}+\frac{c_{p'p}}{4}\right)\alpha_1 \\
\tilde{c}_4 &= \frac{1}{2}\left(c_F-\frac{c_S}{2}\right)
\end{aligned}
$$

Comparison between Lagrangian and On-Shell Formalism-4

Gauge invariance:

$$c_2 = c_4 = 1$$

RPI:

$$2c_F - c_S = 1$$
$$c_F - c_{p'p} = 1$$

Comparison between Lagrangian and On-Shell Formalism-4

Gauge invariance:

$$c_2 = c_4 = 1$$

RPI:

$$2c_F - c_S = 1$$
$$c_F - c_{p'p} = 1$$

Still 2 constraints missing! Why :((

$$c_{W_2} = c_{W_1} - 1$$
$$2c_M = c_F - c_D$$

Off-Shell Current

An off-shell term:

$$
\begin{aligned}
-c_D g \frac{v_\mu \left(D_{\perp\nu} G^{\mu\nu}\right)}{8m^2} &= -\frac{c_D}{8m^2} v_\mu \left[D_\nu G^{\mu\nu} - v_\nu (v \cdot D) G^{\mu\nu}\right] \\
&= -\frac{c_D}{8m^2}\, v \cdot J
\end{aligned}
$$

$$
J^\mu \equiv D_\nu G^{\mu\nu}
$$

Use the current to construct vertices!

Off-Shell 3-pt Vertex

2 equally massive spin-$\frac{1}{2}$ and a current:

$$V_{3v,J} = \bar{u}_{v2} f(v, k_1, k_2, J, m) u_{v1}$$

Off-Shell 3-pt Vertex

2 equally massive spin-$\frac{1}{2}$ and a current:

$$V_{3v,J} = \bar{u}_{v2} f(v, k_1, k_2, J, m) u_{v1}$$

Any matrix sandwiched by heavy quark fields can be represented as [Mannel, 1994]

$$\bar{\psi}_v \Gamma \psi_v = C_I \bar{\psi}_v \psi_v + C_{s,\mu} \bar{\psi}_v s^\mu \psi_v,$$

where $s^\mu = \gamma^\mu \gamma^5$.

Off-Shell 3-pt Vertex-2

Maximally possible vertex:

$$\begin{aligned} V_{3v,J} = &\ (A_1\, v\cdot J + A_2\, k_1\cdot J)\, \bar{u}_{v2} u_{v1} \\ &+ [\epsilon_{\mu\nu\rho\sigma}\, (B_1 J^\nu k_1^\rho v^\sigma + B_2 J^\nu q^\rho v^\sigma) \\ &+ (C_1 J_\mu + C_2\, v\cdot J k_{1\mu} + C_3\, v\cdot J q_\mu)]\, \bar{u}_{v2} s^\mu u_{v1} \end{aligned}$$

$$\begin{aligned} \bar{u}_{v2} u_{v1} = &\ -2\langle 2_v 1_v\rangle \\ \bar{u}_{v2}\gamma^\mu\gamma^5 u_{v1} = &\ 2v^\mu\langle 2_v 1_v\rangle - 2\langle 2_v|\sigma^\mu|1_v] \end{aligned}$$

Relation between Approaches

	On-Shell Spinors	HQET Spinors	HQET Lagrangian
On-Shell	$\langle 21 \rangle$ $\langle 2q \rangle \langle q1 \rangle$	$\langle 2_v 1_v \rangle$ $\langle 2_v \| \frac{k_1}{m} \| 1_v]$ $[2_v \| \frac{k_2}{m} \| 1_v \rangle$ $\langle 2_v \| \frac{k_2 k_1}{m^2} \| 1_v \rangle$	6 operators
Off-Shell	$\langle 23 \rangle \langle 41 \rangle$ $\langle 24 \rangle \langle 31 \rangle$	$v \cdot J \langle 2_v 1_v \rangle$ $k_1 \cdot J \langle 2_v 1_v \rangle$ $\epsilon(J, k_1, v, \langle 2_v \| \sigma \| 1_v])$ $\epsilon(J, q, v, \langle 2_v \| \sigma \| 1_v])$	4 operators

On-shell spinors $\leftrightarrow$ HQET spinors with RPI
RPI in HQET spinors $\leftrightarrow$ RPI in HQET action

Summary and Outlook

- Reparametrization invariance is a non-linearly realized Lorentz symmetry

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Summary and Outlook

- Reparametrization invariance is a non-linearly realized Lorentz symmetry
- Reparametrization invariance relates different orders
- Spinor variables simplify the calculation
- Heavy quark spinors may also be useful in other effective theories explicitly breaking Lorentz symmetry.

Thank you for your attention!